Exercise Sheet

Lecture 10

The Spectral Theorem

Inner products are in the physics convention; the adjoint satisfies $\langle T^\dagger u, v\rangle = \langle u, Tv\rangle$. A $\star$ marks a harder problem.

Diagonalization in an orthonormal basis

Exercise 10.1 (computational) Diagonalize the Hermitian matrix $A = \left(\begin{smallmatrix} 1 & 2 \\ 2 & 1 \end{smallmatrix}\right)$ in an orthonormal basis: find its eigenvalues, an orthonormal eigenbasis, and the unitary U with $U^\dagger AU$ diagonal.

Exercise 10.2 (computational) Show that $A = \left(\begin{smallmatrix} 0 & -2 \\ 2 & 0 \end{smallmatrix}\right)$ is normal but not self-adjoint, and find a unitary diagonalization.

Exercise 10.3 (computational) Diagonalize the real symmetric matrix $A = \left(\begin{smallmatrix} 3 & 1 \\ 1 & 3 \end{smallmatrix}\right)$ in an orthonormal basis: give the eigenvalues, an orthonormal eigenbasis, and the orthogonal O with $O^\top AO$ diagonal.

Exercise 10.4 (computational) Unitarily diagonalize the Hermitian matrix $A = \left(\begin{smallmatrix} 1 & 2i \\ -2i & 1 \end{smallmatrix}\right)$: find its (real) eigenvalues, an orthonormal eigenbasis, and the unitary U with $U^\dagger AU$ diagonal.

Exercise 10.5 (computational) Show that $A = \left(\begin{smallmatrix} 2 & -3 \\ 3 & 2 \end{smallmatrix}\right)$ is normal but not self-adjoint, and give a unitary diagonalization.

Exercise 10.6 (computational) Diagonalize the real symmetric $A = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 3 & 1 \\ 0 & 1 & 3 \end{pmatrix}$ on $\mathbb{R}^3$ in an orthonormal basis, and give $O^\top AO$.

Exercise 10.7 (proof) In the proof of the complex spectral theorem, T is normal and upper-triangular in an orthonormal basis $e_1, \dots, e_n$. Show the first row off-diagonal entries $a_{12}, \dots, a_{1n}$ vanish.

Exercise 10.8 (proof) Let T be normal on a complex inner product space. Using the spectral theorem, prove that T is self-adjoint if and only if every eigenvalue of T is real.

Exercise 10.9 (computational) Diagonalize the Hermitian $A = \left(\begin{smallmatrix} 3 & 1-i \\ 1+i & 2 \end{smallmatrix}\right)$ from scratch: form the characteristic equation, find the (real) eigenvalues, and give an orthonormal eigenbasis and the unitary U with $U^\dagger AU$ diagonal.

Exercise 10.10 (computational) Diagonalize the real symmetric $A = \begin{pmatrix} 14 & -13 & 8 \\ -13 & 14 & 8 \\ 8 & 8 & -7 \end{pmatrix}$ in an orthonormal basis, and give the orthogonal O with $O^\top AO$ diagonal.

Spectral decomposition

Exercise 10.11 (computational) Write the spectral decomposition $A = \lambda_1 P_1 + \lambda_2 P_2$ of $\sigma_z = \mathrm{diag}(1, -1)$, and verify $P_1 + P_2 = I$ and $P_1 P_2 = 0$.

Exercise 10.12 (computational) Using the spectral form, compute $f(A)$ for $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ and $f(\lambda) = \lambda^{1/2}$.

Exercise 10.13 (computational) Write the spectral decomposition $A = \lambda_1 P_1 + \lambda_2 P_2$ of $A = \left(\begin{smallmatrix} 4 & 2 \\ 2 & 4 \end{smallmatrix}\right)$, giving the projectors explicitly, and verify $P_1 + P_2 = I$ and $P_1 P_2 = 0$.

Exercise 10.14 (computational) Using the spectral form, compute the inverse A^{-1} of $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$.

Exercise 10.15 (computational) Find the positive square root $\sqrt{A}$ of the positive operator $A = \left(\begin{smallmatrix} 10 & 6 \\ 6 & 10 \end{smallmatrix}\right)$ via the spectral form.

Exercise 10.16 (computational) Reconstruct the self-adjoint operator A on $\mathbb{C}^2$ with eigenvalue 3 on $\frac{1}{\sqrt{2}}(1, 1)$ and eigenvalue 7 on $\frac{1}{\sqrt{2}}(1, -1)$.

Exercise 10.17 (proof) Prove that an orthogonal projection P is self-adjoint and idempotent ($P^\dagger = P$, $P^2 = P$), and conversely.

Exercise 10.18 (proof) Let T be normal with distinct eigenvalues $\lambda_1, \dots, \lambda_m$ and spectral projections P_k. Prove the interpolation formula $P_k = \prod_{j \neq k} \dfrac{T - \lambda_j I}{\lambda_k - \lambda_j}$.

Measurement and commuting operators

Exercise 10.19 (computational) Measure σ_z on $|\psi\rangle = \frac{1}{\sqrt{10}}(|0\rangle + 3|1\rangle)$. Find the probabilities of ± 1 and the expectation $\langle \sigma_z \rangle$.

Exercise 10.20 ($\star$) Prove that two commuting self-adjoint operators A, B have a common orthonormal eigenbasis.

Exercise 10.21 (computational) On $\mathbb{C}^3$, let $A = \mathrm{diag}(1, 1, 4)$. Find the orthogonal projection onto the 1-eigenspace and the probability that measuring A on $|\psi\rangle = \frac{1}{\sqrt{3}}(1, 1, 1)$ yields 1.

Exercise 10.22 (computational) Measure σ_z on $|\psi\rangle = \frac{1}{2}(\sqrt{3}\,|0\rangle + |1\rangle)$. Find the probabilities of ± 1 and the expectation $\langle \sigma_z \rangle$.

Exercise 10.23 (computational) Measure $A = \mathrm{diag}(1, 2, 3)$ on $|\psi\rangle = \frac{1}{\sqrt{6}}(1, 1, 2)$. Find the probabilities of the three outcomes and $\langle A \rangle$.

Exercise 10.24 (computational) On $\mathbb{C}^3$ let $A = \mathrm{diag}(2, 2, 5)$. Measure A on $|\psi\rangle = \frac{1}{3}(2, 1, 2)$: find the probabilities of each outcome, $\langle A \rangle$, and the post-measurement state when the outcome is 2.

Exercise 10.25 (computational) Verify that $A = \left(\begin{smallmatrix} 1 & 2 \\ 2 & 1 \end{smallmatrix}\right)$ and $B = \left(\begin{smallmatrix} 4 & 1 \\ 1 & 4 \end{smallmatrix}\right)$ commute, and simultaneously diagonalize them in one orthonormal basis.

Exercise 10.26 ($\star$) Suppose T is normal with $T^3 = T$. Show that the eigenvalues of T lie in $\{0, 1, -1\}$ and that T is self-adjoint.

Exercise 10.27 (proof) For an observable $A = \sum_k \lambda_k P_k$ with the spectral projections P_k, prove that the Born-rule probabilities $\|P_k |\psi\rangle\|^2$ of a normalized state $|\psi\rangle$ sum to 1.

Perturbation theory

Throughout, $H(\varepsilon) = H_0 + \varepsilon V$ with H_0 and V self-adjoint and $\varepsilon \in \mathbb{R}$, and denominators are written with the level of interest first.

Exercise 10.28 (computational) Let $H_0 = \mathrm{diag}(1,4)$ and $V = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$. Find the first- and second-order shifts of both levels, and compare the second-order prediction with the exact eigenvalues at $\varepsilon = 0.1$.

Exercise 10.29 (computational) For $H_0 = \mathrm{diag}(0,2,5)$ and

$$V = \begin{pmatrix} 1 & 1 & 0 \\ 1 & -1 & 2 \\ 0 & 2 & 0 \end{pmatrix},$$

compute all three first- and second-order shifts, and verify the two checks $\sum_n E_n^{(1)} = \operatorname{tr} V$ and $\sum_n E_n^{(2)} = 0$.

Exercise 10.30 (computational) For the three-level system of the lecture, $H_0 = \mathrm{diag}(0,1,3)$ with V having rows $(2,1,1)$, $(1,1,1)$, $(1,1,-3)$, compute the first-order correction to the *top* eigenvector and evaluate it at $\varepsilon = 0.1$.

Exercise 10.31 (computational) Let $H_0 = \mathrm{diag}(0,1,3)$ and $V = \left(\begin{smallmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{smallmatrix}\right)$. Compute $\|V\|$, and find the largest $|\varepsilon|$ for which the three disks of the spectral stability theorem are still pairwise disjoint.

Exercise 10.32 (proof) *(Hellmann–Feynman.)* Let $|n(\varepsilon)\rangle$ be a normalized eigenvector of $H(\varepsilon)$ depending differentiably on ε, with eigenvalue $E_n(\varepsilon)$. Prove that

$$\frac{dE_n}{d\varepsilon} = \langle n(\varepsilon)|\, V \,|n(\varepsilon)\rangle,$$

and deduce the first-order formula $E_n^{(1)} = \langle n|\, V \,|n\rangle$.

Exercise 10.33 (proof) Assume H_0 has simple eigenvalues. Prove that $\sum_n E_n^{(2)} = 0$, and that the second-order shift of the *lowest* level is always ≤ 0 while that of the highest is always ≥ 0.

Exercise 10.34 (computational) Let $H_0 = \mathrm{diag}(2,2,6)$ and

$$V = \begin{pmatrix} 1 & 3 & 0 \\ 3 & 1 & 1 \\ 0 & 1 & 0 \end{pmatrix}.$$

Explain why the non-degenerate formulas cannot be used, then find the first-order splitting of the degenerate level and the correct zeroth-order states.

Exercise 10.35 ($\star$) For $\Delta > 0$ consider the two-level family

$$H(\varepsilon) = \begin{pmatrix} \Delta/2 & \varepsilon \\ \varepsilon & -\Delta/2 \end{pmatrix}.$$

(a) Find the exact eigenvalues. (b) Expand the upper one to order ε^2 and check it against second-order perturbation theory. (c) Regarding ε as a complex variable, find the radius of convergence of that expansion, and say what obstructs it.